SURVIVAL OF BENEFICIAL ENCAPSULATED STRAIN *LACTOCOCCUS LACTIS* MK1/3 IN LACTOSE-FREE COW MILK YOGURT

Natália Zábolyová, Eva Bino, Monika Pogány Simonová, Andrea Lauková

Centre of Biosciences of the SAS, Institute of Animal Physiology, 040 01 Košice

Introduction

Lactose intolerance, a common condition caused by a lack of the lactase enzyme, leads to digestive discomfort after consuming dairy. Functional foods, such as lactose-free products enriched with beneficial bacteria such as *Lactococcus lactis*, help manage this condition by producing β-galactosidase, which acts similarly to lactase (Ishikawa et al., 2015). *Lactococcus lactis* strain MK1/3 (our strain Accession number in GenBank-ON114093), isolated from raw goat milk, is safe and produces postbiotic substance (PS) of proteinaceous character with antimicrobial properties (Lauková et al., 2025a).

Aim of the Study

This study aims to evaluate the stability and survival of encapsulated *Lactococcus lactis* MK1/3 in lactose-free cow milk yogurts.

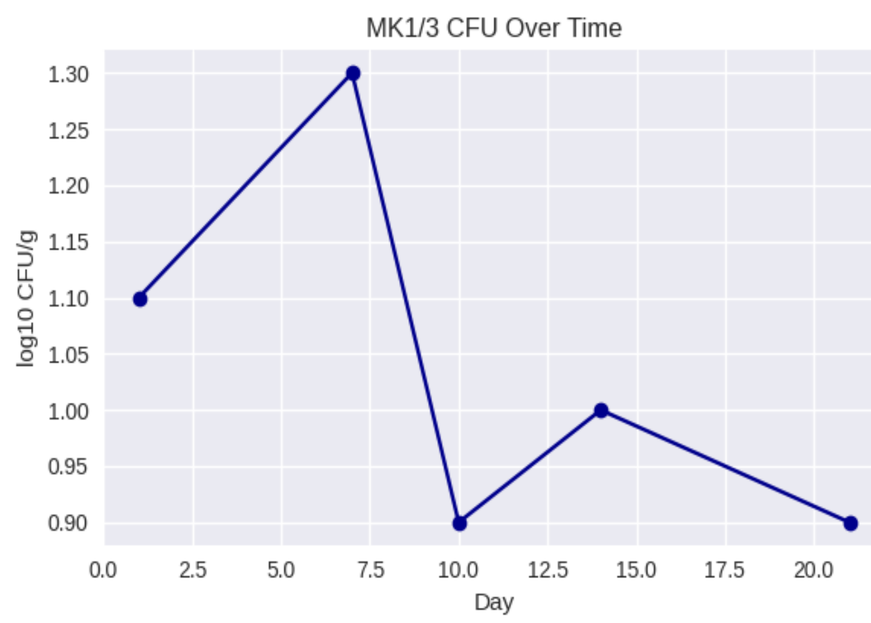
Materials & Methods

- **Encapsulation**: *Lactococcus lactis* MK1/3 was freeze-dried after growth in M17 broth and mixed with skim milk
- **Samples**: Lactose-free yogurts were inoculated with 0.5 g of encapsulated MK1/3; controls were not.
- **Design**: Samples were taken on days 0/1, 1, 7, 10, and 14.
- **Analyses**: MK1/3 survival (CFU/g) and pH were measured. All tests were in duplicate.

Results

Survival of *Lactococcus lactis* MK1/3

Lactococci counts increased from 4.84 ± 0.21 CFU/g log 10 at 24 h to 10.10 ± 0.00 CFU/g at day 21 in experimental yogurts, and from 4.80 to 10.60 ± 0.25 CFU/g in controls.



Lactococci Growth

Lactococci counts increased from 4.84 ± 0.21 CFU/g at 24 h to 10.10 ± 0.00 CFU/g at day 21 in experimental yogurts, and from 4.80 to 10.60 ± 0.25 CFU/g in controls.

pH Stability

pH remained stable between 3.20 and 3.25 throughout the study in both experimental and control yogurts.

Strain Properties & Patent

MK1/3 forms a desirable coagulate, has optimal fermenting ability, and is included in the patent (PP 50021-2022, patent number 289266). It shows potential for lactose-free yogurt fortification and as a β-galactosidase producer for lactose-intolerant people.

Conclusion

The safe strain L. lactis MK1/3, isolated from raw goat milk and producing β-galactosidase, was encapsulated and added to lactose-free cow milk yogurts. It survived in the highest count on day 7, with total lactococci increasing in both experimental and control yogurts. The pH remained stable throughout the study. Although MK1/3 did not reach high counts, it shows potential for further fortification in lactose-free products after additional testing.

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Zábolyová Natália MVDr.
kotesovska@saske.sk

Lauková Andrea, MVDr., Csc.
laukova@saske.sk

INSTITUTE OF ANIMAL PHYSIOLOGY

CENTRE OF BIOSCIENCES SLOVAK ACADEMY OF SCIENCES